

Hamza Mustafa

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Education

Skidmore College

B.A. in Computer Science, Minor in Mathematics; Major GPA: 3.95/4.00

May 2027

Saratoga Springs, NY

Professional Experience

Skidmore College

AI Research Assistant, NVIDIA's IsaacLab (Prof. Wenlu Du, Assistant Professor of CS)

Sep 2024 – Present

Saratoga Springs, NY

- Designed RL algorithms in **Nvidia's Isaac Lab (Linux)** to train Unitree **Robot** for bipedal locomotion across 4,096 parallel environments.
- Optimized model weights and hyper-parameters, **boosting model stability by 75%** and reducing convergence time by 20%.
- Presented "*Bipedal Locomotion in Quadrupedal Robots Using Reinforcement Learning*" to **120+ attendees**.
- **Featured on College Newspaper** for innovative work in AI

ProgressSoft Corporation

Software Developer Intern

Aug – Sep 2025

Amman, Jordan

- Built high-quality backend services designed using Spring Boot, supporting reliable data retrieval for financial transactions.
- Implemented REST APIs and optimized database queries, improving system responsiveness and stability.
- Built a JavaScript (React) dashboard for real-time data visualization and asset monitoring.
- Containerized services with Docker to support scalable internal deployment and integration with existing systems.

Tamatem Games

Software & Legal Intern

May – Aug 2024

Amman, Jordan

- Coordinated with QA, DevOps, and engineering teams during Agile sprints to validate features, identify defects, and track development progress.
- Authored a business analysis report with actionable insights that would increase annual profits by 5%.
- Supported cross-functional communication between technical, legal, and publishing teams in a fast-paced mobile-gaming environment.

Projects

High-Performance Financial Transactions Platform (Spring Boot, React)

- **Improved financial data retrieval performance by 35%** by designing a full-stack CRUD platform using Spring Boot REST APIs, optimized SQL queries, and a React frontend.

Self-Learning Checkers AI (Java, Reinforcement Learning)

- **Outperformed a traditional alpha-beta pruning agent** by implementing a SARSA-based reinforcement learning model for competitive game-play.
- **Enhanced AI system usability and observability** by visualizing game-play and training metrics using a Spring Boot WebSocket backend and React front end.

AI Assistant That Answers Questions About Your Favorite Philosophers (Python, Lang-chain, Llama)

- **Enabled accurate, context-aware question answering over custom philosophical sources** by building a RAG pipeline using Llama 3.2's API and embedding-based retrieval.

Machine Learning Model That Detects Cancer from Medical Data (Python, Tensor flow)

- **Achieved 92% accuracy in tumor malignancy classification** by preprocessing medical datasets with Pandas and training a TensorFlow model using Sigmoid activation, Binary Cross-Entropy loss, and Adam optimization.

Additional Experience

Skidmore College

Learning Assistant — Design & Analysis of Algorithms (CS305), Visiting Asst. Prof. Esma Wali

Sep 2025 – Present

Saratoga Springs, NY

- **Tutored** and provided algorithmic feedback for high level algorithms class, reinforcing rigor in problem-solving and analysis.

Skidmore College

Peer Academic Coach Captain (Computer Science)

Jan 2025 – Present

Saratoga Springs, NY

- **Trained** coaches to support **computer science students**, hosting **Leetcode** sessions and tutoring students.

Technical Skills

Languages: Java, Python, C, GoLang, SQL, Javascript, R

Frameworks/Tools: Omniverse, IsaacLab, Spring Boot, React, PyTorch, TensorFlow, NumPy, Pandas, Docker, Git, Maven, Postman, Unity, Arduino **Concepts:** Reinforcement Learning, Machine Learning, API Design, Database Modeling, Data Visualization/Tableau